Gas Flow Simulation in L-PBF Chamber

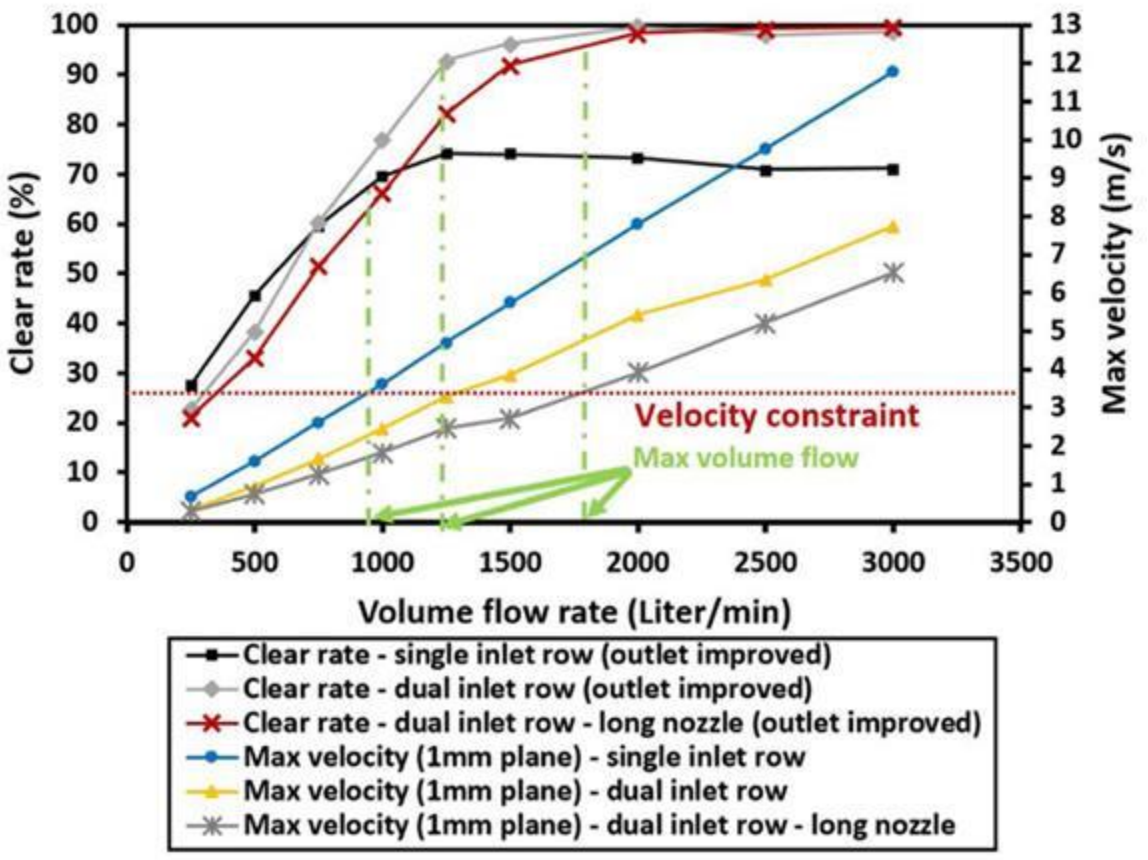
Contents

Requirements for gas flow in the chamber

- O2 content of lower than 1000 or 500 ppm
- Pumping time: 1 hour
- Clear rate = $\left(1-\frac{N_r}{N_t}\right) * 100\%$

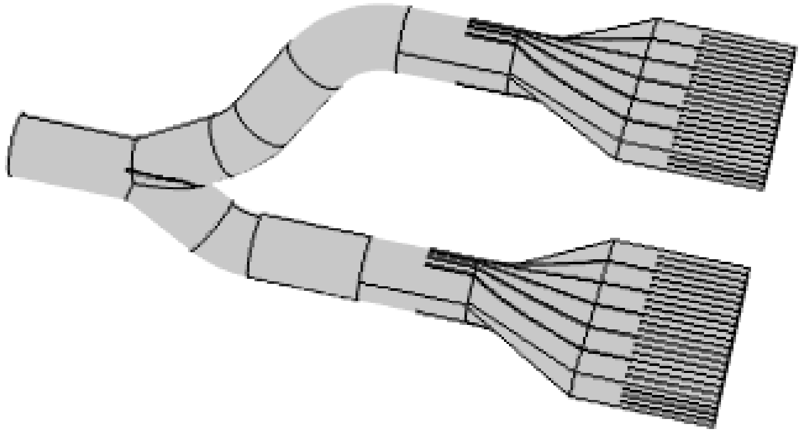
Where N_r is the re-deposited (trapped) spatters in build region (powder bed), N_t is the total number of generated spatters.

- Maximun clear rate: 95%
- The threshold velocity at 1mm above powder bed can be computed= 3.383 m/s

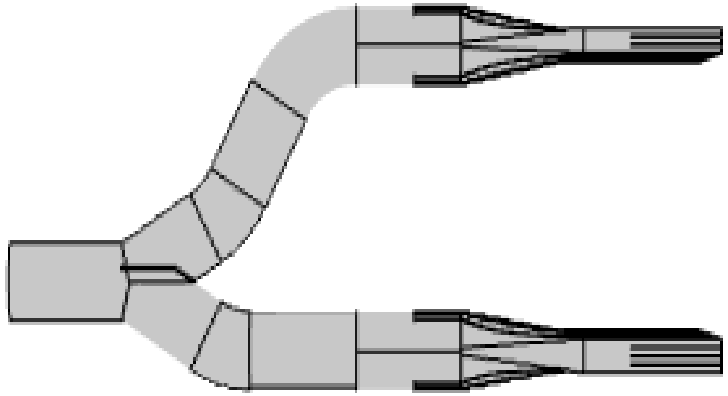


1. Chamber Design

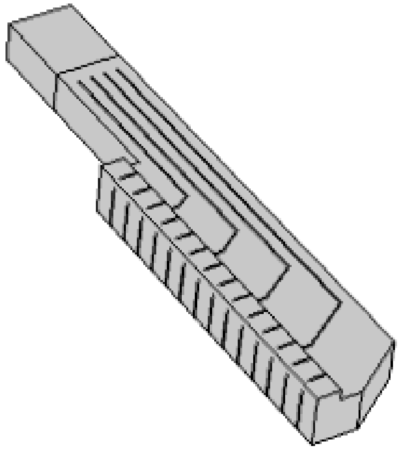
- Design of inlet



- Design of inlet Guide vanes

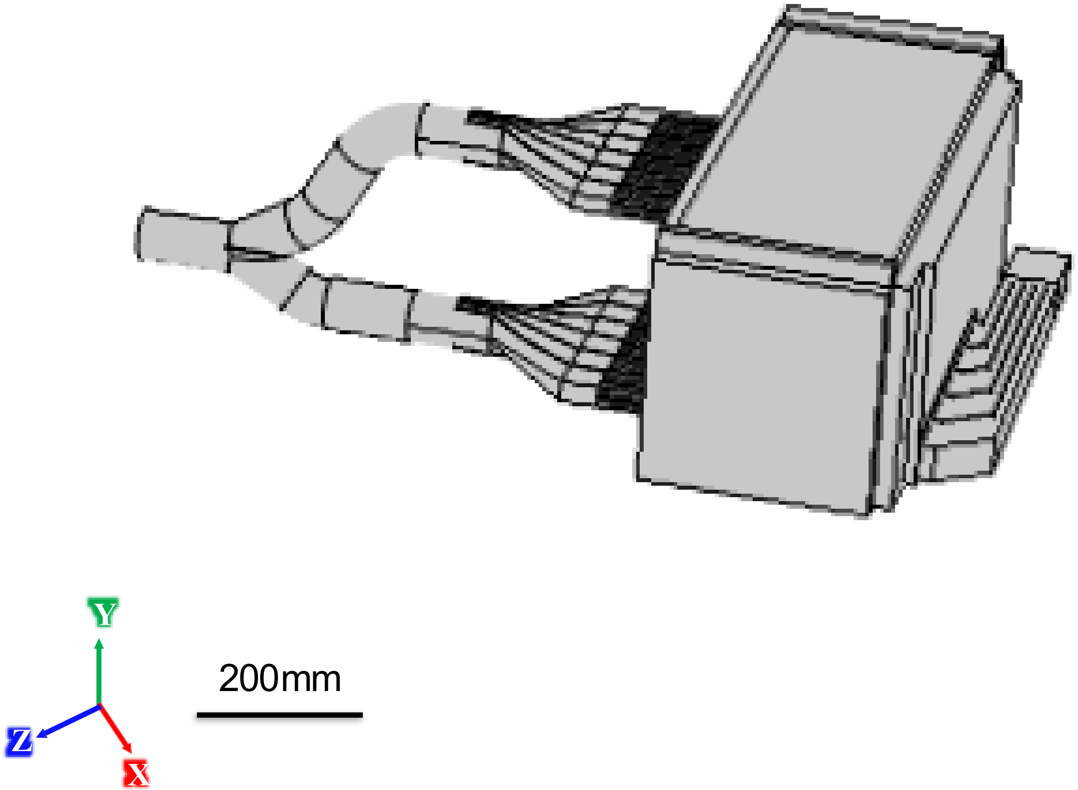


- Design of outlet



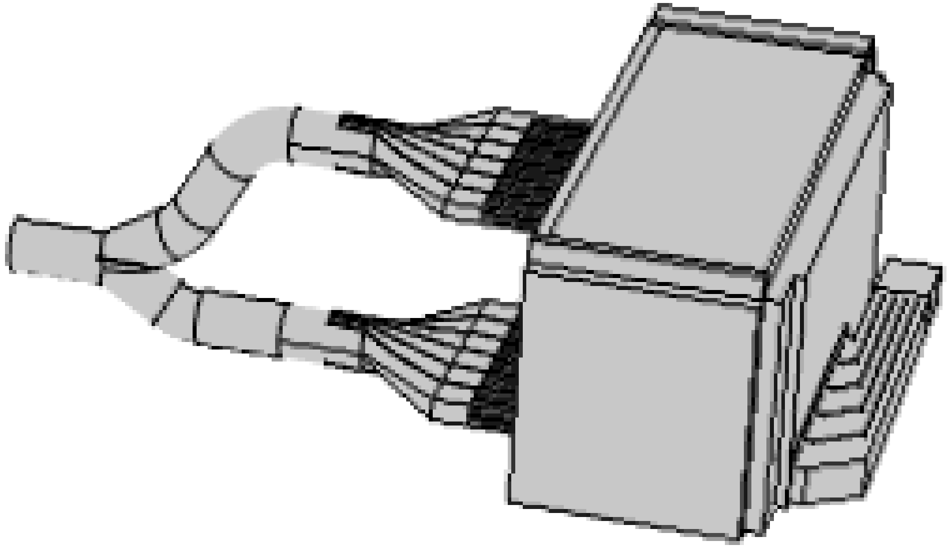
1. Chamber Design

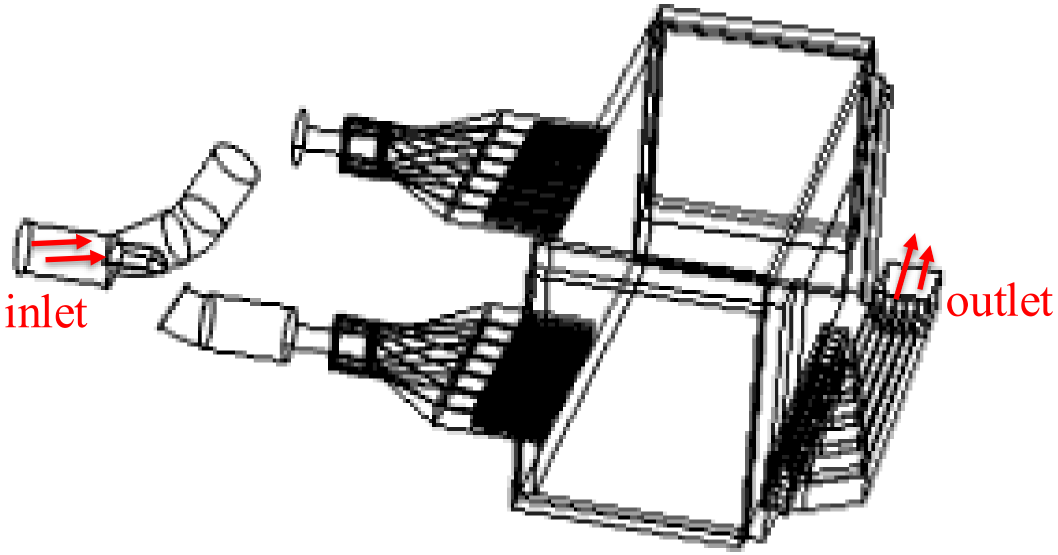
- Design of Chamber

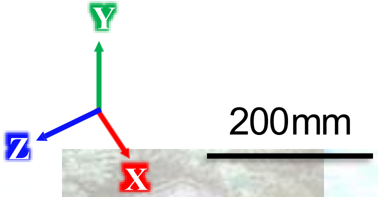


2. Mesh Configuration / Boundary condition setup

- Inlet Flow Velocity 20 m/s
- Flow rate: 0.183 m^3/s

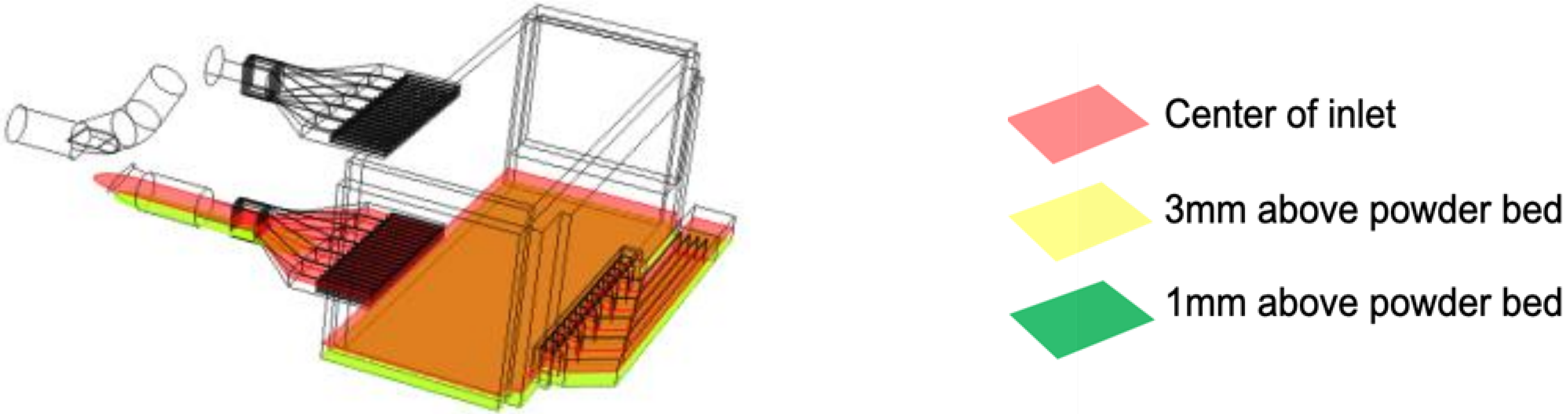






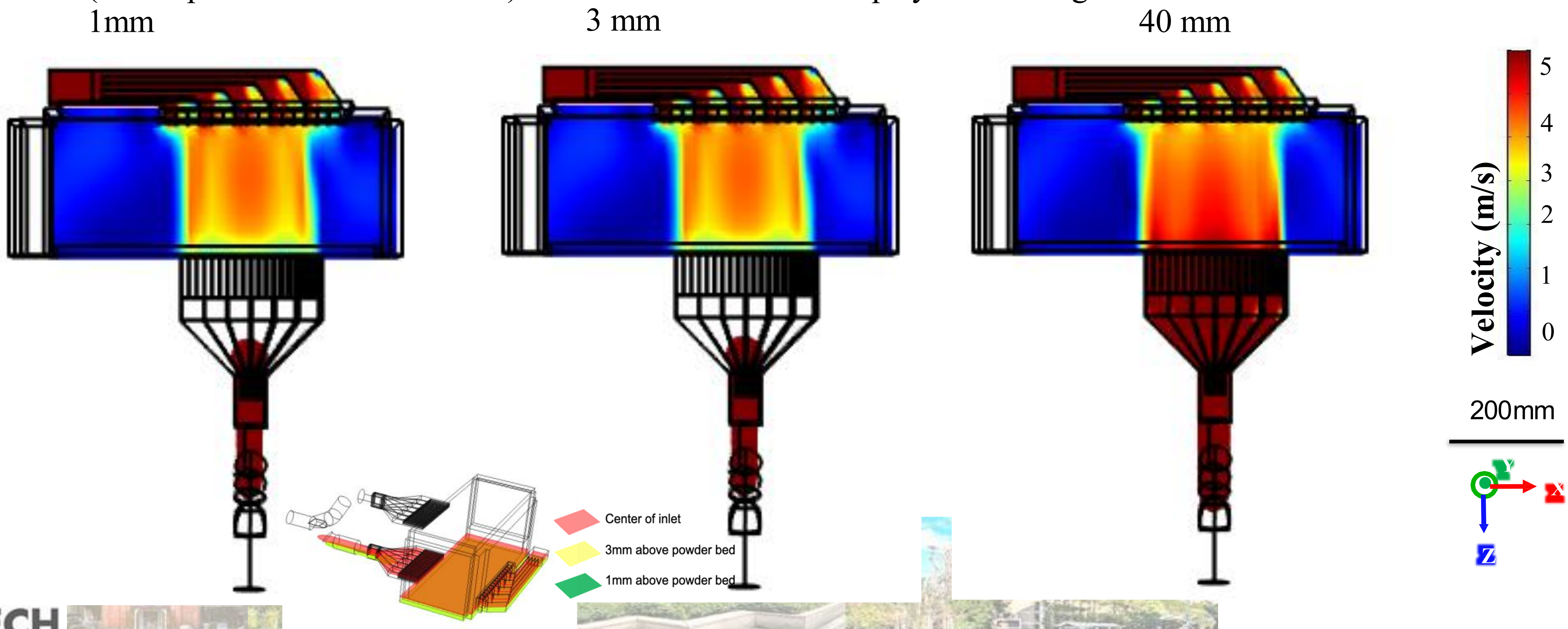
Simulation results

Base Plane



Simulation results

- The simulated results of velocity magnitude distribution on a horizontal plane above the powder bed at distances of 1 mm, 3 mm, and 40 mm (center plane of the inlet nozzle) were extracted and are displayed in the figure below.



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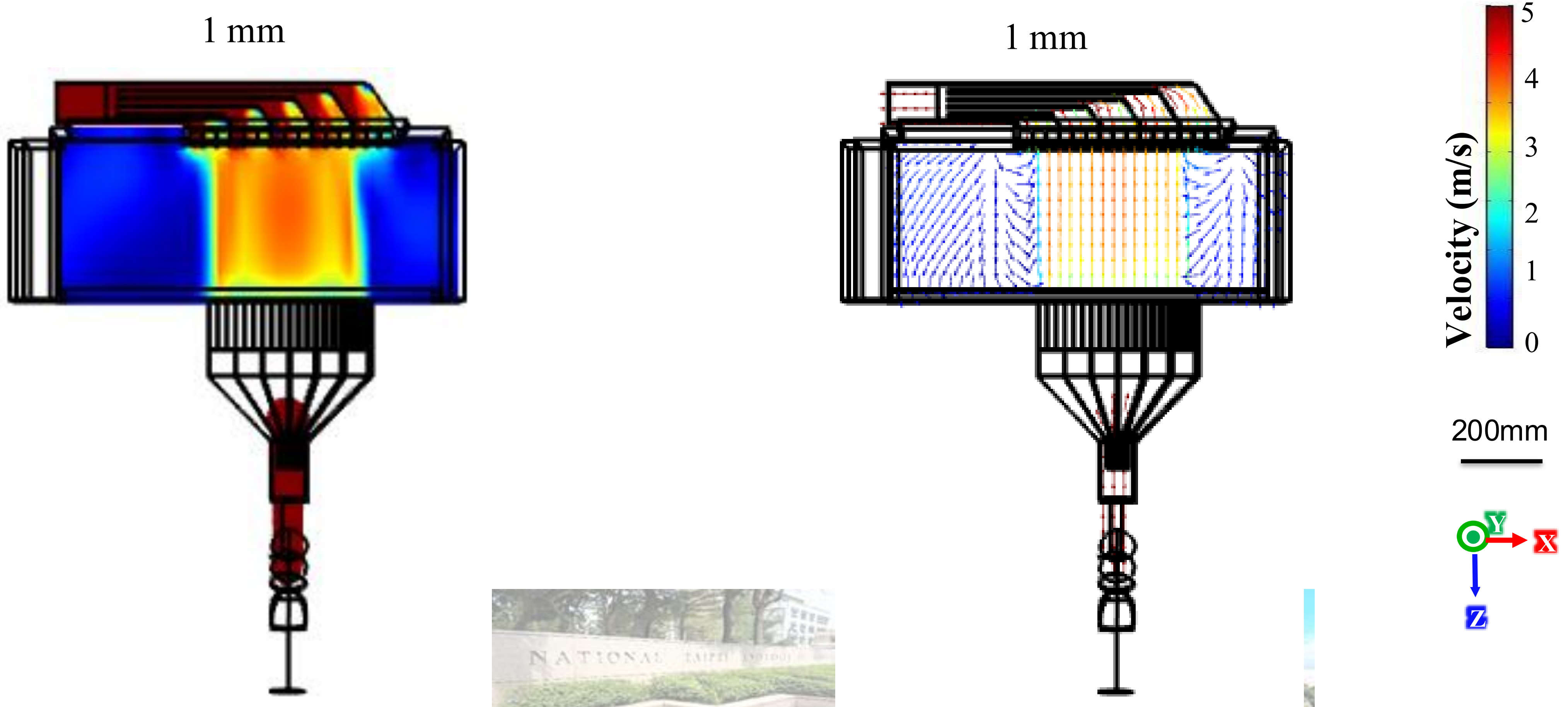
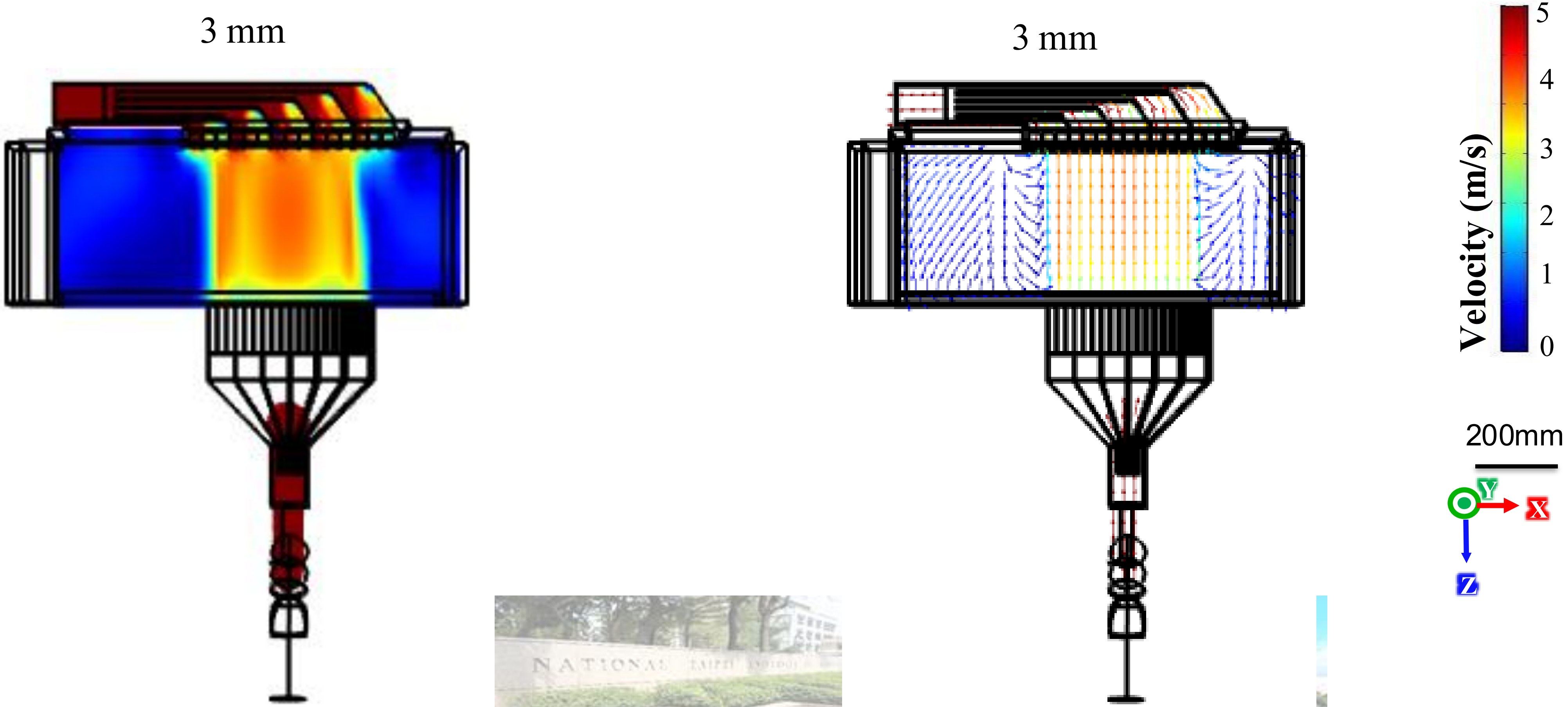
Simulation results

- The flow field velocity vectors distribution on a horizontal plane above the powder bed at distances of 1 mm are shown in Figure below
- Velocity magnitude distribution on a vertical cross-section



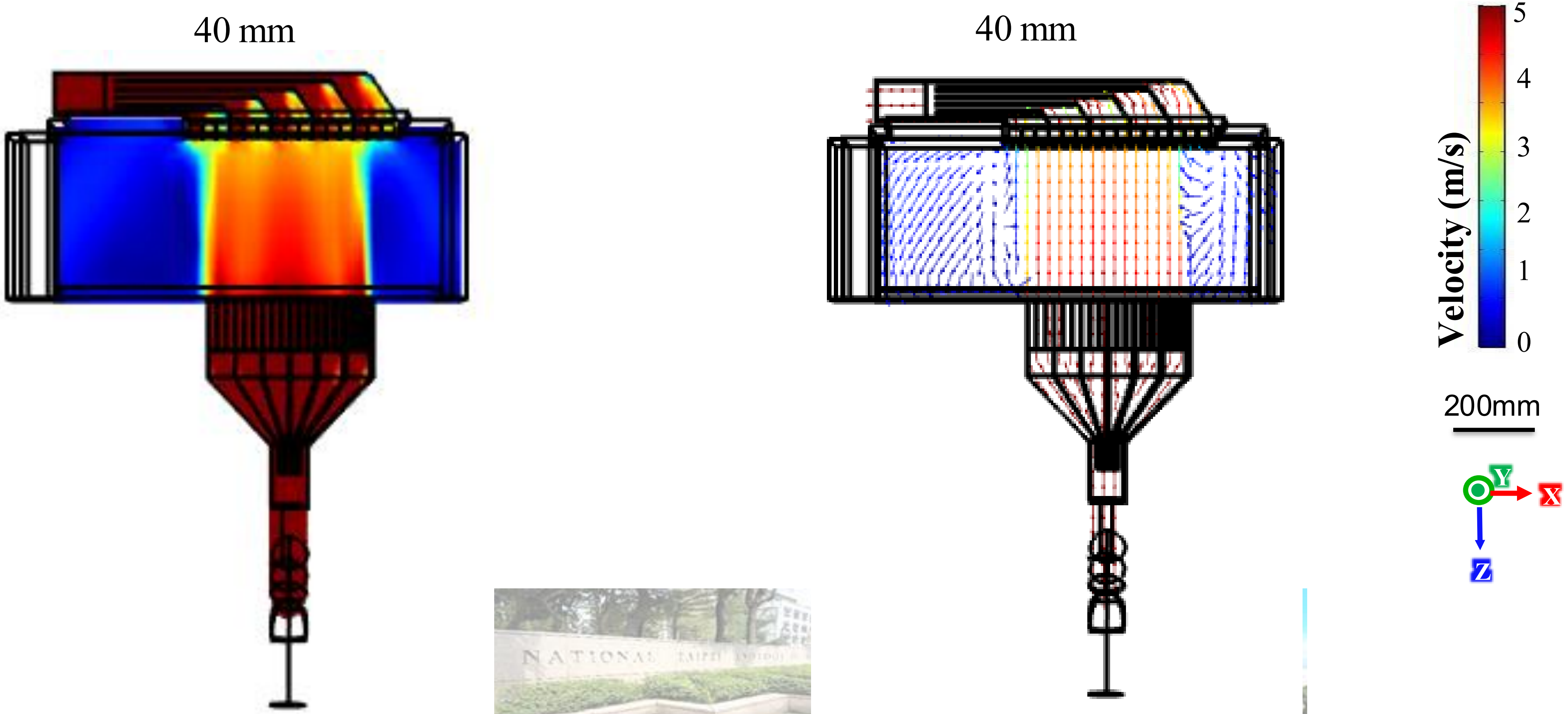
Simulation results

- The flow field velocity vectors distribution on a horizontal plane above the powder bed at distances of 3 mm are shown in Figure below
- Velocity magnitude distribution on a vertical cross-section



Simulation results

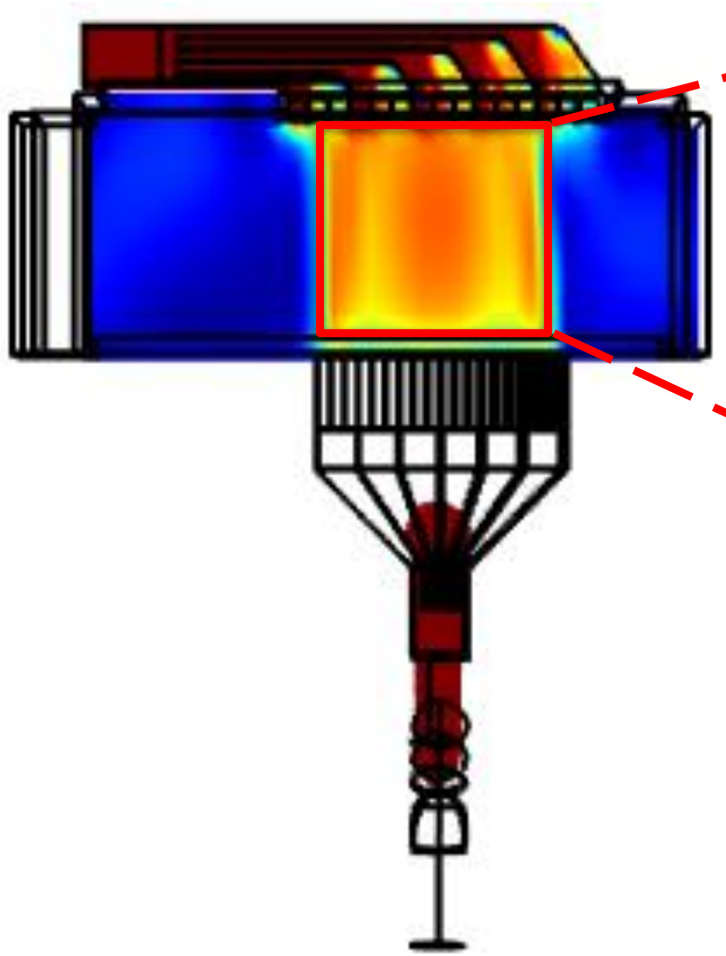
- The flow field velocity vectors distribution on a horizontal plane above the powder bed at distances of 40 mm are shown in Figure below
- Velocity magnitude distribution on a vertical cross-section

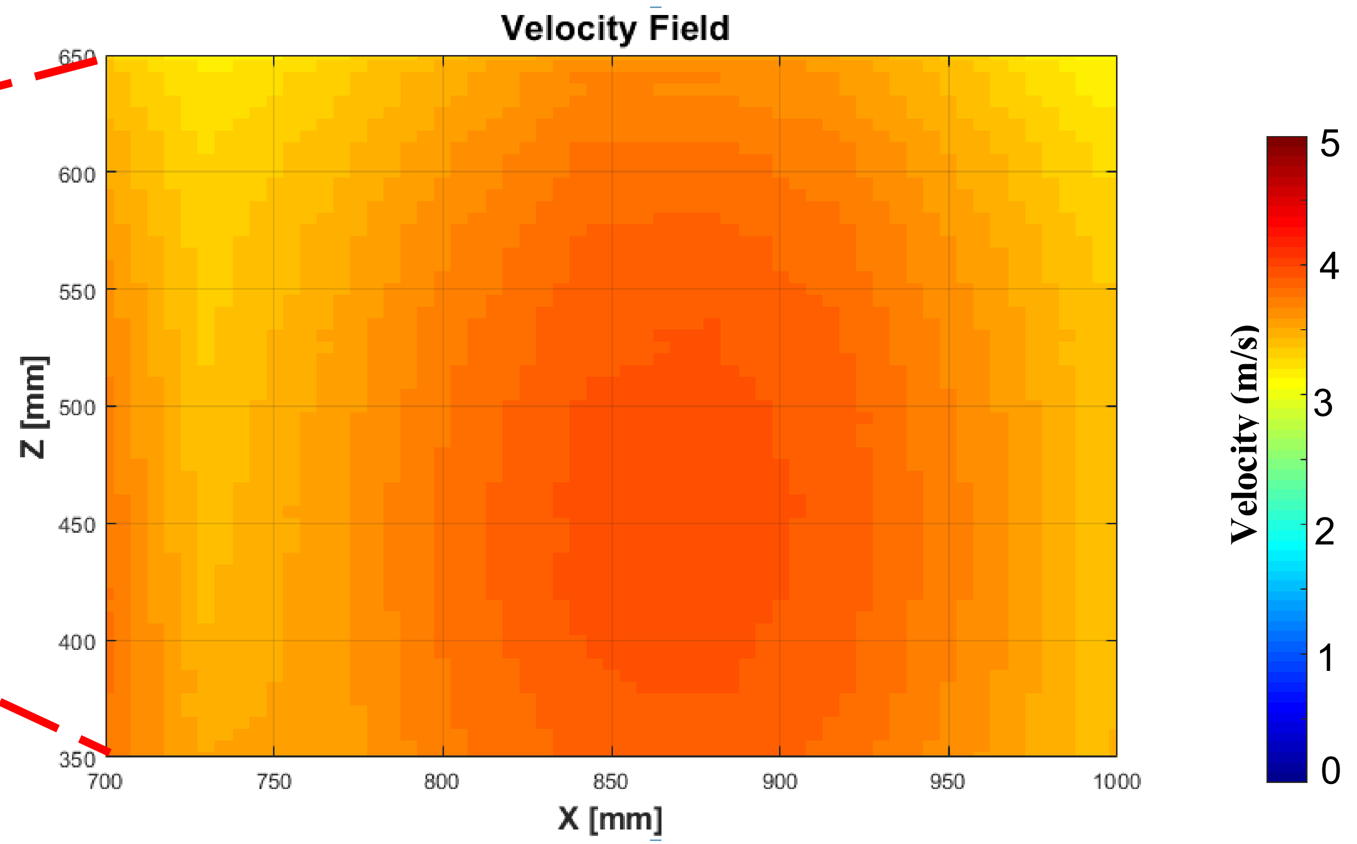


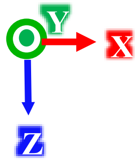
Simulation Results

Extract Location :

- Base Plane : Above powder bed 1 mm
- Top View



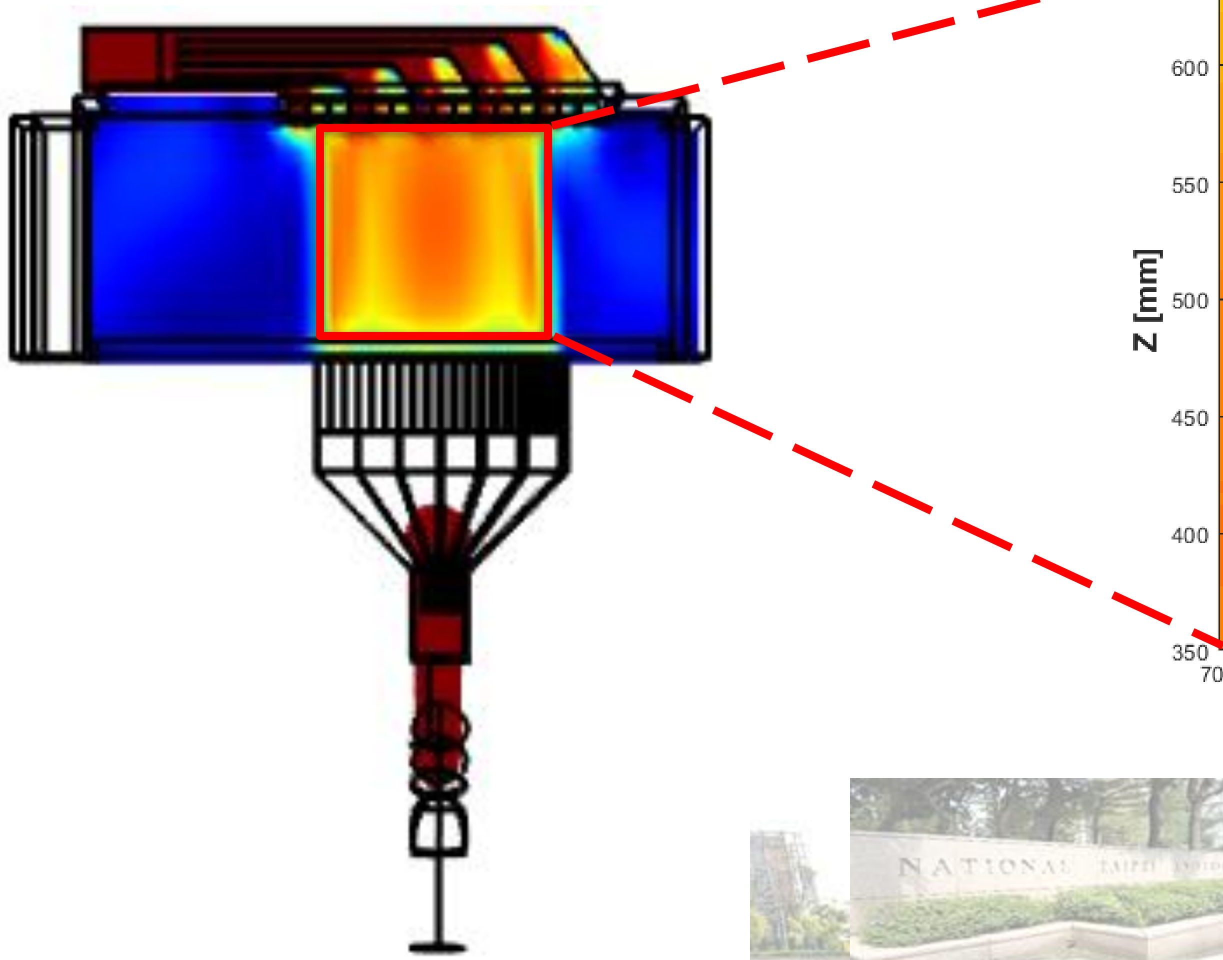


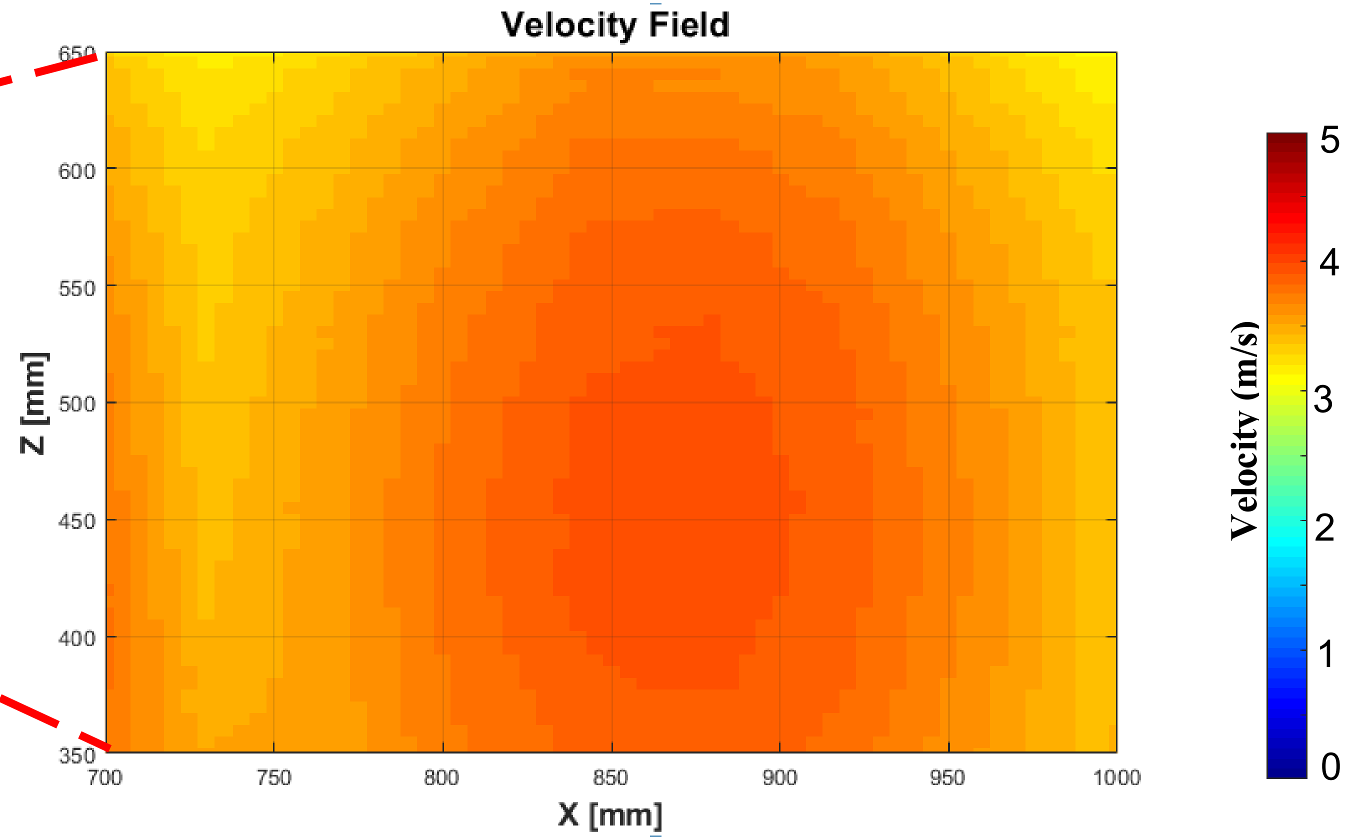


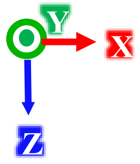
Simulation Results

Extract Location :

- Base Plane : Above powder bed 3 mm
- Top View



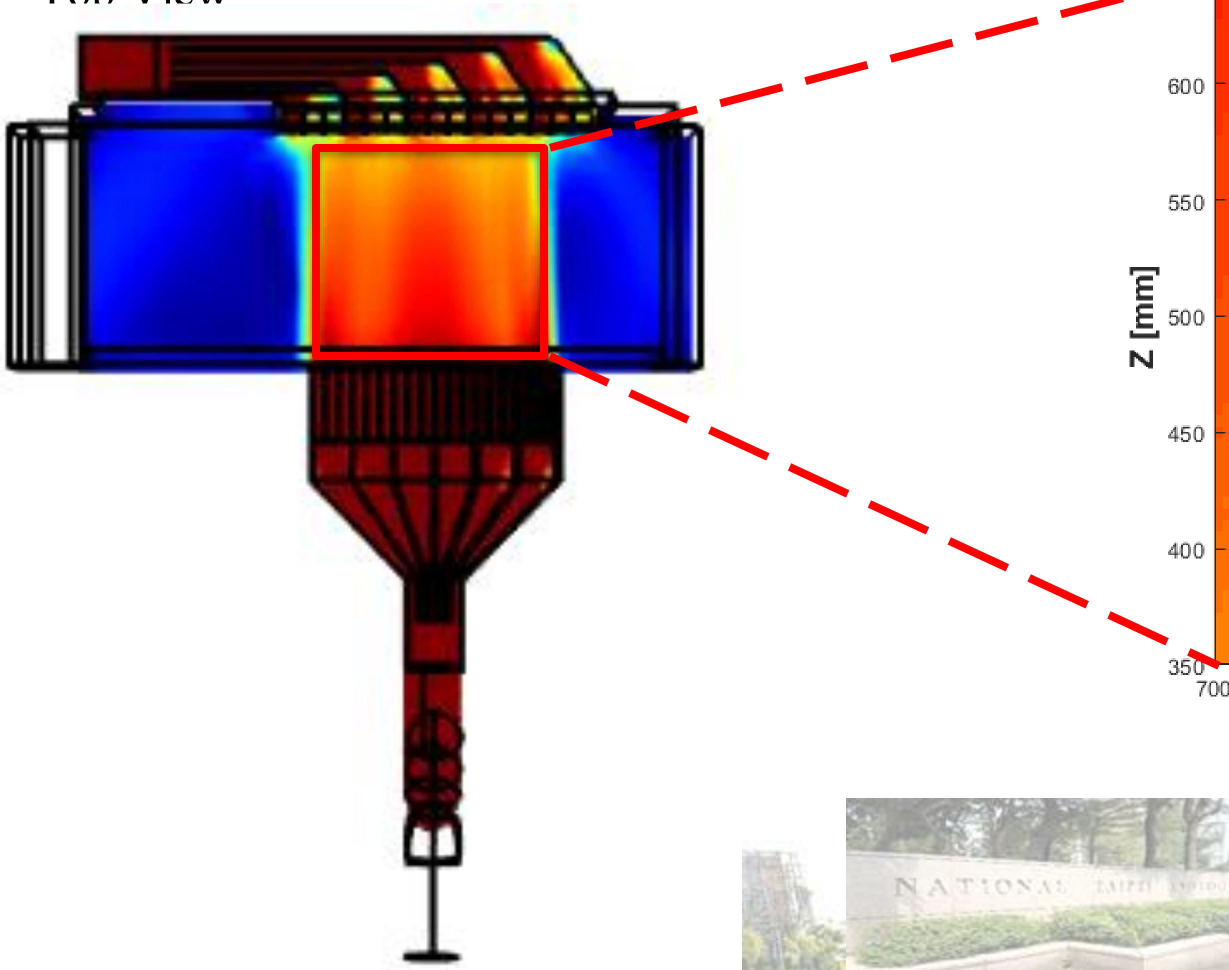


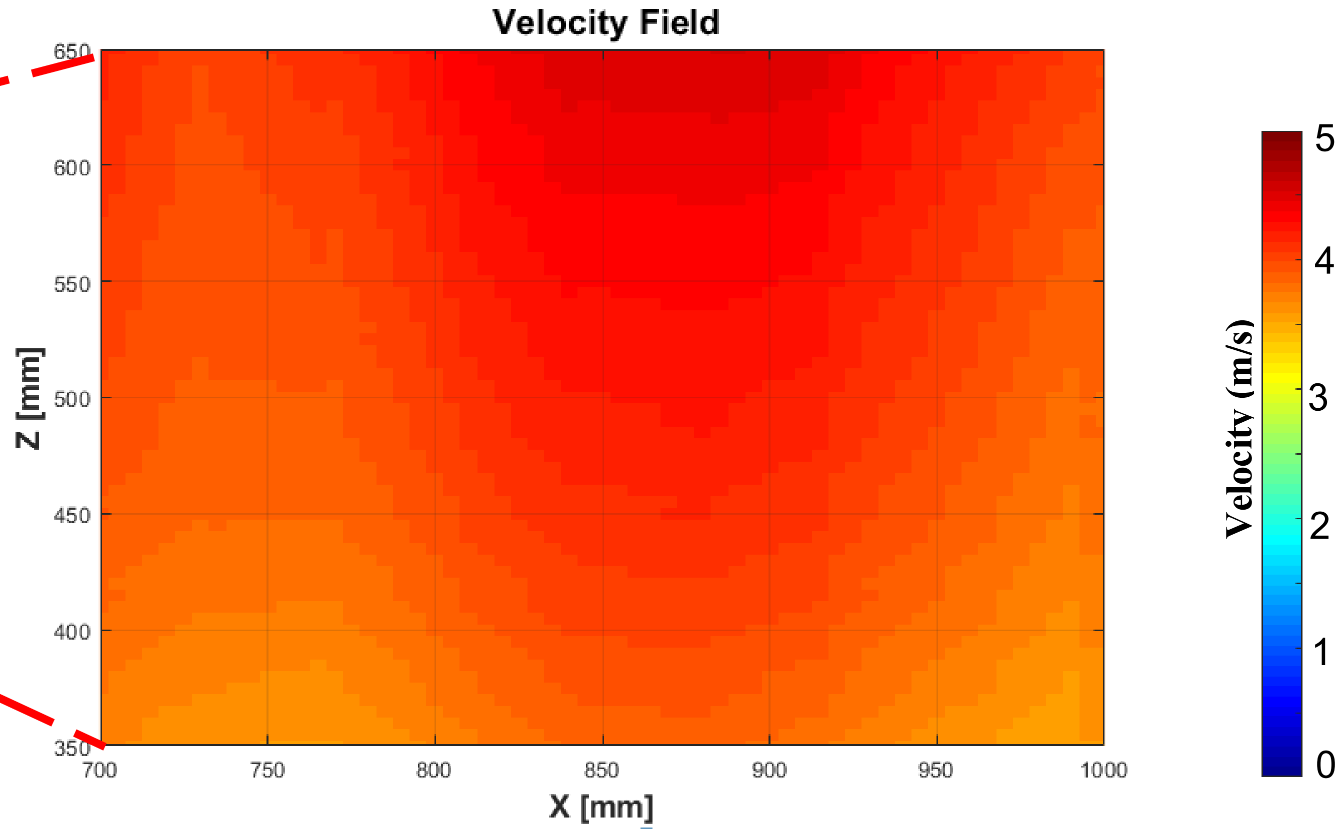


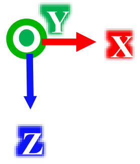
Simulation Results

Extract Location：

- Base Plane：Above powder bed 40 mm
- Top View



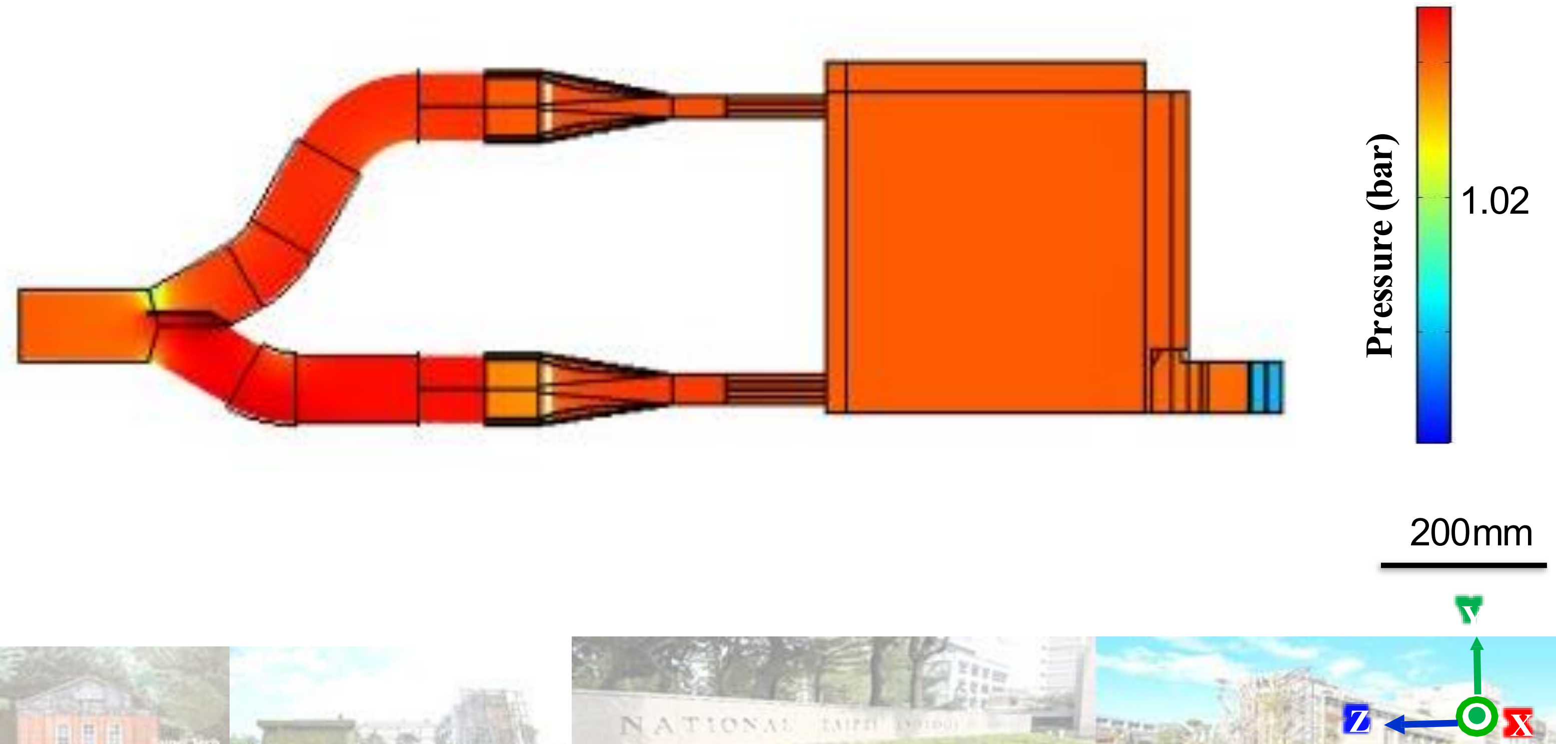




Simulation Results

Extract Location :

- Base Plane yz : Center of inlet
- Side View
- Pressure distribution
- Pressure drop: 0.012 bar



Simulation Results

Velocity Distribution Comparison